

Disproof of conjecture 00000009523: the completely dephasing qubit channel has $C_E = \chi = \log 2$

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: The entanglement-assisted classical capacity $C_E(\varphi)$ is given by the Bennett-Shor-Smolín-Thapliyal formula $C_E(\varphi) = \max_{\rho} I(\rho, \varphi)$, where $I(\rho, \varphi) = S(\rho) + S(\varphi(\rho)) - S((\text{id} \otimes \varphi)(\text{purification}))$ is the quantum mutual information. Conjecture: Except for the completely depolarizing constant channel, $C_E(\varphi) > \chi(\varphi)$ strictly for every channel; and the maximal ratio C_E/χ over d -dimensional channels is d , attained by trace-halving channels, with the attaining channel unique up to joint unitary equivalence. (The Chinese text states the same.)

Reading. The conjecture is a conjunction. Its first conjunct is the universal claim

$$(A) \quad \forall \text{ channels } \varphi \neq \Delta : \quad C_E(\varphi) > \chi(\varphi),$$

where $\Delta(X) = \text{tr}(X) I/d$ is the completely depolarizing channel. We refute (A) with $d = 2$, which refutes the conjunction. We do not address the second conjunct (ratio C_E/χ), which is not needed. The witness is non-constant on states and has $\chi > 0$, so it also refutes (A) under two weaker readings: (A') every channel that is constant on states is excepted, and (A'') in addition only channels with $\chi(\varphi) > 0$ are considered.

Conventions. Natural logarithm (the claim $C_E = \chi$ does not depend on the base). $\eta(x) = -x \log x$ with $\eta(0) = 0$ (Mathlib `Real.negMulLog`). States are on $\mathbb{C}^d$; the purification of a state on $\mathbb{C}^d$ lives in $\mathbb{C}_R^d \otimes \mathbb{C}_A^d$ with the reference R as the *first* factor, and φ acts on the second factor. The supremum in C_E is taken over all states and all purifications of each state (we show it is attained, so it is the maximum).

2 Definitions (as formalized)

Definition 1. A *state* is a matrix $\rho \in M_d(\mathbb{C})$ that is positive semidefinite with $\text{tr } \rho = 1$. The *von Neumann entropy* of a Hermitian matrix A is $S(A) = \sum_i \eta(\lambda_i)$, where λ_i are the eigenvalues of A with multiplicity (Mathlib `Matrix.IsHermitian.eigenvalues`).

Definition 2. For $X \in M_{k \cdot n}(\mathbb{C})$ indexed by pairs (r, a) , and a linear map $\Phi : M_n \rightarrow M_m$, $(\text{id}_k \otimes \Phi)(X)_{(r,a),(r',a')} = \Phi(X_{rr'})_{aa'}$, where $X_{rr'}$ is the block $(a, b) \mapsto X_{(r,a),(r',b)}$. A *channel* is a $\mathbb{C}$ -linear map Φ that is completely positive ($(\text{id}_k \otimes \Phi)(X) \succeq 0$ for every k and every $X \succeq 0$) and trace preserving.

Definition 3. For $\psi \in \mathbb{C}^d \otimes \mathbb{C}^d$, $|\psi\rangle\langle\psi|$ is the matrix $\psi\psi^*$, and tr_R is the partial trace over the first factor. ψ is a *purification* of ρ if $\text{tr}_R |\psi\rangle\langle\psi| = \rho$. The mutual information is $I(\rho, \psi, \Phi) = S(\rho) + S(\Phi\rho) - S((\text{id} \otimes \Phi)|\psi\rangle\langle\psi|)$ and $C_E(\Phi) = \sup\{I(\rho, \psi, \Phi) : \rho \text{ a state, } \psi \text{ a purification of } \rho\}$. This

is the formula of Bennett, Shor, Smolin and Thapliyal, whose abstract (arXiv:quant-ph/0106052) describes the capacity as “the maximum, over channel inputs ρ , of the entropy of the channel input plus the entropy of the channel output minus their joint entropy, the latter being defined as the entropy of an entangled purification of ρ after half of it has passed through the channel.”

Definition 4. The *Holevo quantity* of a finite ensemble ($p_i \geq 0$, $\sum_i p_i = 1$, states ρ_i) is $S(\sum_i p_i \Phi(\rho_i)) - \sum_i p_i S(\Phi(\rho_i))$, and the *Holevo capacity* $\chi(\Phi)$ is its supremum over all finite ensembles.

The witness is the *completely dephasing qubit channel* $D(X) = \text{diag}(X_{00}, X_{11})$ on $M_2(\mathbb{C})$.

3 Proof

Lemma 5 (entropy from the characteristic polynomial). *If A is Hermitian and $\det(tI - A) = \prod_i (t - w_i)$ with $w_i \in \mathbb{R}$, then $S(A) = \sum_i \eta(w_i)$. In particular $S(\text{diag}(w)) = \sum_i \eta(w_i)$.*

Proof. The roots of the characteristic polynomial, with multiplicity, are the eigenvalues (`IsHermitian.roots_charpoly_eq_eigenvalues`); they are also the multiset $\{w_i\}$. Apply $\eta \circ \text{Re}$ and sum. $\square$

Lemma 6. *Every qubit state ρ has $S(\rho) \leq \log 2$.*

Proof. The two eigenvalues satisfy $\lambda_0 + \lambda_1 = \text{tr } \rho = 1$, so $S(\rho) = \eta(\lambda_0) + \eta(1 - \lambda_0) = h(\lambda_0) \leq \log 2$, where h is the binary entropy (`Mathlib.Real.binEntropy_le_log_two`). $\square$

Lemma 7. *D is a channel, it is not constant on states (it fixes $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$), and hence $D \neq \Delta$.*

Proof. Trace preservation is clear. With the diagonal projectors $P_a = \text{diag}(\mathbf{1}[\text{second index} = a])$ on $\mathbb{C}^k \otimes \mathbb{C}^2$ one checks entrywise that $(\text{id}_k \otimes D)(X) = \sum_{a \in \{0,1\}} P_a^* X P_a$, a sum of positive semidefinite matrices when $X \succeq 0$. Finally $\Delta(\rho) = I/2$ for every state. $\square$

Lemma 8 (key identity). *For every $\psi \in \mathbb{C}^2 \otimes \mathbb{C}^2$, with $\rho = \text{tr}_R |\psi\rangle\langle\psi|$, we have $I(\rho, \psi, D) = S(\rho)$.*

Proof. Let $\tau_a = \sum_r |\psi(r, a)|^2$. Then $D(\rho) = \text{diag}(\tau_0, \tau_1)$, so $S(D\rho) = \eta(\tau_0) + \eta(\tau_1)$ by Lemma 5. Entrywise, $\omega := (\text{id} \otimes D)|\psi\rangle\langle\psi|$ is the block-diagonal matrix $\text{blockDiag}_a(v_a v_a^*)$ with $v_a = \psi(\cdot, a) \in \mathbb{C}^2$. Its characteristic polynomial is $\prod_a \det(tI - v_a v_a^*) = \prod_a t(t - \tau_a)$ (a 2×2 rank-one matrix has determinant 0 and trace τ_a). By Lemma 5, $S(\omega) = \eta(\tau_0) + \eta(\tau_1)$. Hence $S(D\rho) = S(\omega)$ and $I = S(\rho)$. $\square$

Theorem 9. $C_E(D) = \log 2 = \chi(D)$, both suprema are attained, and $\chi(D) > 0$.

Proof. C_E : by Lemmas 8 and 6, every $I(\rho, \psi, D) \leq \log 2$. For $\psi = (|00\rangle + |11\rangle)/\sqrt{2}$, $\rho = I/2$ and $I = S(I/2) = 2\eta(1/2) = \log 2$.

χ : for any ensemble, write $d_{i,a} = (\rho_i)_{aa} \in [0, 1]$ (real since $\rho_i \succeq 0$), $d_{i,0} + d_{i,1} = 1$. Then $\sum_i p_i D(\rho_i) = \text{diag}(\sum_i p_i d_{i,a})$, so the Holevo quantity is $\sum_a \eta(\sum_i p_i d_{i,a}) - \sum_i p_i \sum_a \eta(d_{i,a}) \leq \log 2 - 0$, using the binary-entropy bound and $\eta \geq 0$ on $[0, 1]$. The ensemble $\{\frac{1}{2} : |0\rangle\langle 0|, \frac{1}{2} : |1\rangle\langle 1|\}$ gives $S(I/2) - 0 = \log 2$. $\square$

Theorem 10. *Clause (A) is false for $d = 2$: D is a channel, $D \neq \Delta$, and $C_E(D) = \chi(D)$, so $C_E(D) > \chi(D)$ fails. Readings (A') and (A'') also fail with the same witness.*

This agrees with the literature: Shirokov (arXiv:1202.3449) writes “It is easy to show coincidence of the entanglement-assisted classical capacity and the Holevo capacity for any c-q channel between finite dimensional quantum systems.” The channel D is a c-q channel (measure in the computational basis, prepare the corresponding basis state). The Lean proof does not use this result.

4 Lean formalization and axiom audit

File `lean/Conjecture9523/Basic.lean` (namespace `C9523`, Lean v4.33.1, Mathlib v4.33.1, 338 lines, only import `Mathlib`). Definitions: `IsState`, `vnEntropy`, `proj`, `ptraceRef`, `idTensor`, `IsChannel`, `mutualInfo`, `mutualInfoSet`, `CE`, `holevoQ`, `holevoSet`, `chi`, `depol`, `deph`. Main lemmas: `vnEntropy_of_charpoly` (Lemma 5), `vnEntropy_le_log_two` (Lemma 6), `deph_isChannel`, `deph_nonconstant`, `deph_ne_depol` (Lemma 7), `idTensor_deph_proj`, `charpoly_blockDiagonal`, `vnEntropy_omega`, `mutualInfo_deph` (Lemma 8), `CE_deph`, `chi_deph` (Theorem 9, each as `IsGreatest` of the defining set plus the value of `sSup`).

Main theorems:

```
theorem deph_counterexample :
  IsChannel deph /\ deph != depol 2 /\
    (exists rho sigma,
      IsState rho /\ IsState sigma /\ deph rho != deph sigma) /\
      0 < chi deph /\ CE deph = chi deph /\ chi deph = log 2
theorem conjecture9523_false :
  not (forall Phi : Matrix (Fin 2) (Fin 2) C -> l[C] Matrix (Fin 2) (Fin 2) C,
    IsChannel Phi -> Phi != depol 2 -> chi Phi < CE Phi)
theorem conjecture9523_false_strong :
  not (forall Phi : Matrix (Fin 2) (Fin 2) C -> l[C] Matrix (Fin 2) (Fin 2) C,
    IsChannel Phi ->
      (exists rho sigma,
        IsState rho /\ IsState sigma /\ Phi rho != Phi sigma) ->
        0 < chi Phi -> chi Phi < CE Phi)
```

(ASCII transcription. The Lean source writes $\wedge$, $\neq$, $\exists$, $\neg$, $\forall$, $\rightarrow$, ρ , σ , Φ , $\mathbb{C}$, and $\rightarrow_l [\mathbb{C}]$ for $\mathbb{C}$ -linear maps.)

Axiom audit. `#print axioms` for each of the three theorems reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide`, and no added axiom. `lake build` succeeds.

Sources

- C. H. Bennett, P. W. Shor, J. A. Smolin, A. V. Thapliyal, Entanglement-assisted capacity of a quantum channel and the reverse Shannon theorem, <https://arxiv.org/abs/quant-ph/0106052>.
- M. E. Shirokov, A criterion for coincidence of the entanglement-assisted classical capacity and the Holevo capacity of a quantum channel, <https://arxiv.org/abs/1202.3449>.